

AI-Assisted Avatar Fashion Show: Word-to-Clothing Texture Exploration and Motion Synthesis for Metaverse UGC

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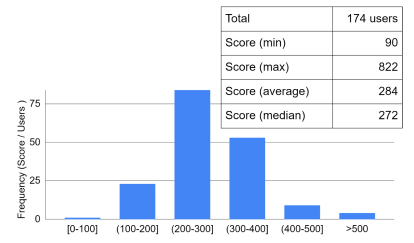


Figure 1: The project’s user experience during field testing: WordToClothing texture generation (Left), motion recording using Muscle-Compression, and choreography. We generated 174 original avatars and their motions in 5 days through a public exhibition (mid). The histogram (right) shows the distribution of the proposed evaluation method for 174 users in the exhibition.

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1 INTRODUCTION

Developing new user-generated content in the metaverse, a communication space with 3D avatars, can enhance the user experience and value of computer graphics, interactive techniques, and their applications [Ball 2022; Berthault et al. 2023].

This study, AI-Assisted Avatar Fashion Show (AAAFS), focuses on user-generated content (UGC), which is increasingly important in the metaverse. It proposes a solution to produce valuable UGC while wearing an HMD for a broader audience in a short time. This AI-assisted method is expected to not only give the newness of using HMD devices, but also respect the user’s creativity and guarantee originality, in comparison to the conventional smartphone of PC metaverse.

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2 SYSTEM ARCHITECTURE OF AAAFS

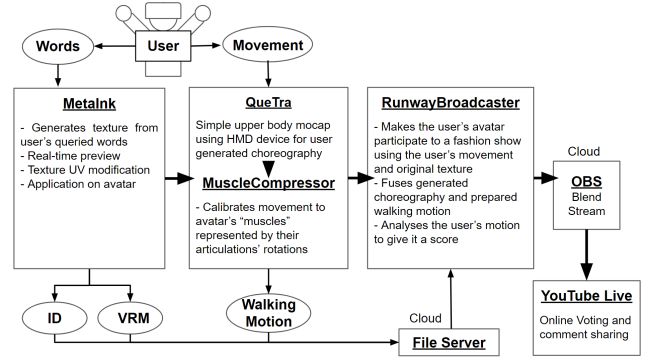


Figure 2: AAAFS system configuration

This AAAFS experience is composed of these three systems, “MetaInk”, “MuscleCompressor” and “RunwayBroadcaster”. These systems reach end user, cloud and YouTube audiences.

MetaInk. We conceived MetaInk, an app for HMD, to create and personalize the clothing texture. It allows the users to generate their original dress texture from few words queries, and apply this texture to the clothing of avatars in real-time using our WordToClothing technology. We have used the Stable Diffusion 1.5 (v1-5-pruned-emaonly.ckpt) model and added “texture pattern” as a prompt before the given words like “flower”, “cat”, “leopard”, “pink”. The tiling and color of the texture can also be controlled with our shader

with HMD user interfaces in real-time. Modifying a texture's pixels can be a very performance intensive process, in particular when you do operations on the whole texture. Therefore, in our system, operations such as tiling, changing color or applying masks are done through material modification in the process depicted in figure 3. In this system, a camera gets the image of a mask layer applied atop a pattern layer. The mask layer represents the avatar's skin part of the texture. The pattern is the layer that we update with the generated texture. A pencil function is also available to directly modify texture pixels as a paint operation, but WordToClothing, UV adjustment, and color correction shaders can be sufficient to achieve the desired results in a short time (<3min). Once the creations are validated by the user, the avatar as a whole, as a .vrm file, and an associated unique ID are created and sent to our server.

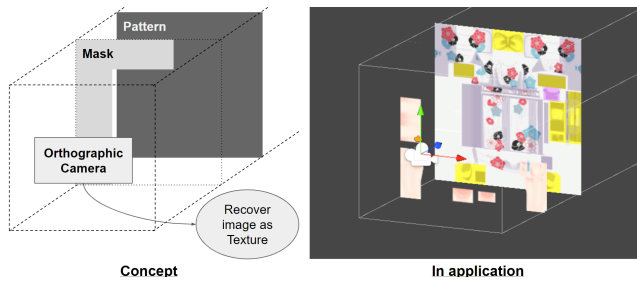


Figure 3: From the orthographic camera we can retrieve the texture with the mask, tiling and color applied.

MuscleCompressor. MuscleCompressor consists of a capture system, QueTra, that records upper body movements using an HMD, and an internal animation generation algorithm that compresses the user motion for later AI-fusion generation. A general way of recording motion, using position (xyz) and rotation (xyzw of Quaternion) of all bone in each frame is not optimized for avatar communications, so iterating on our AI-Fusion system [Kato et al. 2022], we reduced the data volume by regenerating it as Muscle system in Unity. Muscle allows the corresponding bone to move within a range of motion (min, max) normalized to one local axis. In this case, we reduced the data volume by about 86% by serializing the values of 55 body muscles and 40 finger muscles as a list in MotionDataClass and storing them as byte data. MotionDataClass contains information on a single frame, including the global unique identifier (GUID) to identify the motion, the frame's date and time, and the object's name and value recorded in the muscle, which is stored as a list for the number of frames. The saved motions are sent to our server, it takes less than 1 min for 1 min sequence.

RunwayBroadcaster. RunwayBroadcaster uses AI-Fusion technology to create fashion show scenes for YouTube Live with generated avatars and recorded motion. Simultaneously, motion is evaluated using a discrete Fourier transform spectrum with principal frequencies of 1.6 Hz and 3.2 Hz, indicating the pre-recorded lower body animation by a professional motion actor.

$$\text{MotionMatchingScore} = \frac{P_{1.6\text{Hz}} + P_{3.2\text{Hz}}}{\sum_k P_{k\text{Hz}}} \times 1000 \quad (1)$$

This allows users wearing Meta Quest2, a 3-point tracker playing a knocker sound at 96 BPM, to align their upper body motion with pre-recorded lower body animation, ensuring flexible evaluation for different types of motions including back-beat and double-tempo. This method brings motion evaluation closer to human subjectivity, guarantees the quality of motion UGC by flexible playing with ready-made professional motions, and eliminates inconsistencies between devices with different frame rates. It allows for the fusion of quality motion and low-resolution motion (e.g., upper and lower body) in a flexible manner.

3 EVALUATION

The effectiveness of the proposed system was evaluated through field tests in a public exhibition and YouTube live streams. External coordinators were present on site to explain the basic operation of WordToClothing and to supervise the participants in French. The experience was designed to last about 3 minutes until YouTube showing. Three gaming notebooks (NVIDIA® GeForce RTX™ 2080) were used for the entire system, including texture generation, HMD experience, and YouTubeLive. A total of 174 files were collected during the five-day evaluation period, with participants ranging in age from 8 to 60 years, without any duplicated (un-original) results in WordToClothing and/or animation. The experience was comfortable for the users, without complex explication, and was easily understood by the children. As a UX evaluation, users empirically reported feeling stressed when the time from query to display exceeded 30 seconds, which can be improved by using Stable Diffusion fine-tuning, step parameters, texture transfer methods, and/or network environments. As an objective assessment, 99.2% of the users were able to generate their original UGCs. The histogram (Figure 1, right) is obtained as the motion and rhythm matching frequency as scores, as shown in. This method can indicate that this evaluation method successfully classifies and streamed in real-time on YouTube Live, where it could be objectively evaluated through comments as a collective evaluation.

4 CONCLUSION

We have succeeded in creating an attractive technology and experience that encourages a wide diversity of users, in their genders and generations, to wear HMDs and generate high-quality UGC, such as avatar clothing and movement choreography. At the same time, this is a good example of how AI technology can extend creativity and share metaverse experiences that can quickly generate new content for everyone, not only in person but also in online streaming media, without taking away human agency.

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